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ESI-MS in Inorganic Chemistry

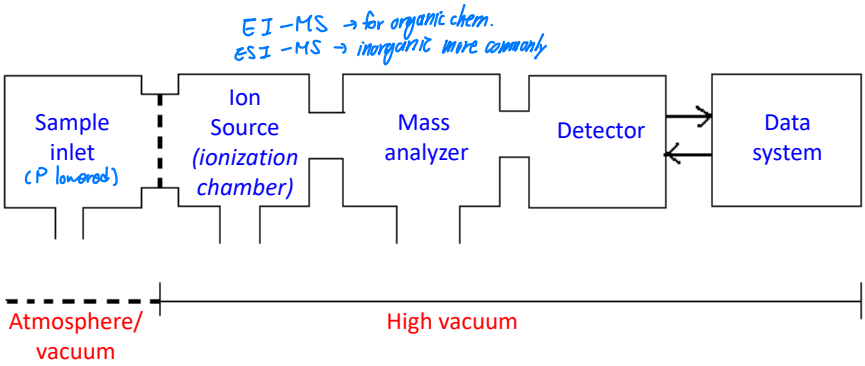
This topic is related to "Experiment 1: Homogenous Catalysis & Experiment 2: Metallocene".

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Mass Spectrometer

- The components of mass spectrometer:



Reference: D. L. Pavia, G. M. Lampman, G. S. Kriz, J. R. Vyvyan, *Introduction to Spectroscopy*, 4th ed.

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Mass Spectrometer

- It has five components in its *simplest form* of mass spectrometer:

(i) sample inlet
- the sample is brought from atmosphere (i.e. lab. condition) to the lower pressure of the instrument;
(ii) ion source / ionization chamber
- the sample molecules are transformed into gas phase ions;
(iii) mass analyzer
- the sample ions are separated according to their mass-to-charge (m/z) ratio;
(iv) detector
- the separated ions are counted;
(v) data system
- the signal is recorded and processed, and a mass spectrum is generated.

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Common Ionization Methods Used in Inorganic Chemistry

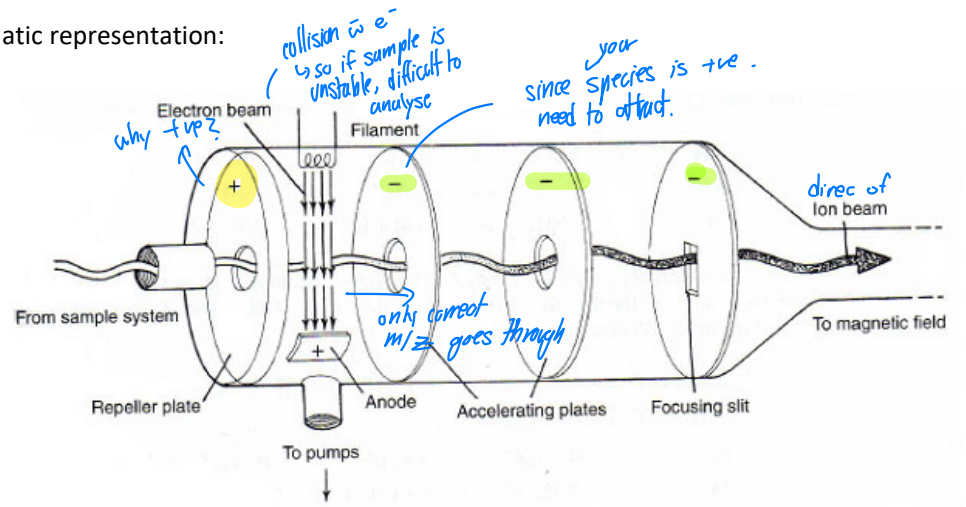
- **Electron ionization (EI)**
  - the simplest and most common way for converting sample to ions;
  - requirement(s): the sample must be relatively volatile

- **Electrospray ionization (ESI)**
  - another type: Thermospray Ionization (TSI) → similar mechanism
  - to study high M.W. biomolecules and small molecules, labile or nonvolatile compounds
  - requirement(s): a solution of sample
  - types of analytes: too nonvolatile, too polar, thermally labile

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Electron Ionization (EI)

Schematic representation:



Source: D. L. Pavia, G. M. Lampman, G. S. Kriz, J. R. Vyvyan, *Introduction to Spectroscopy*, 4th ed..

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Electron Ionization (EI)

- Use high-energy electrons, generated by filament at high temperature, to strike the stream of molecule coming from the sample inlet system;

- Several components in EI chamber:
- (1) filament – being heated to very high temperature to emit high-energy electrons
- (2) repeller plate – directing the newly created ions toward the accelerating plates
- (3) focusing plate / focusing slit – directing the ions into a uniform beam

- Advantages: (i) the hardware is inexpensive and robust, (ii) fragmentation pattern of a compound is reproducible, (iii) libraries of EI-MS data are available.

- Disadvantages: Difficult to analyze the compounds with short lifetime fragments or high molecular weight.

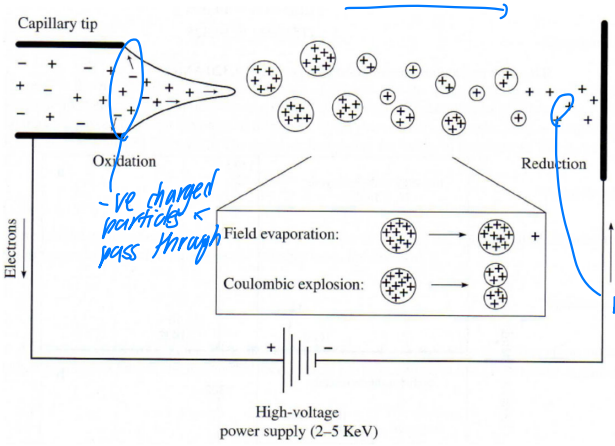
e.g. : ITMS (+p) ESI E Full ms [200.00 -1400.00]
↓ +ve charge
↓ full spec range
NL: 9.75E3
↓ conc.
dep on sample & solvent amount
↳ influence peak height & relative ratio of impurity
↳ can do both +ve & -ve charged species. just the settings diff.

Electrospray ionization (ESI)

no high E e⁻.
↳ no collision
↳ x initiate fragmentation
DH ÷ cf EI-MS

- Schematic representation:

droplet size ↓ due to N_2 gas → higher charge density
↓
repulsion & breakdown to smaller droplet (solvent free)
* no solvent in sample reaching the end.
BUT, may coordinate if sample has metal & solvent donor group.



-ve charged particles pass through

may combine due to high P
↳ get higher m/z than expected
↳ solve by diluting or lowering P

Source: D. L. Pavia, G. M. Lampman, G. S. Kriz, J. R. Vyvyan, *Introduction to Spectroscopy*, 4th ed..

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Electrospray ionization (ESI)

- Processes:

(i) a sample solution is sprayed out the end of a fine capillary into a heated chamber;
(ii) as high voltage potential is applied, small charged droplets are expelled into the ionization chamber;
(iii) an uncharged carrier gas (usually nitrogen) is used to help evaporate the neutral solvent in the droplets;
(iv) finally, solvent-free sample ions generated are left in gas phase for analysis.

- The sample ions may carry a single charge or multiple charges.

*low intensity peaks 看.
for organic, C & H isotopes small proportion + MW quite similar
but for inorganic, halogen, metals etc
get intense peak
more than 1

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Isotope Ratio Data

- With using the relative intensities of the peaks of molecular ion (i.e. M) and related isotope(s) (i.e. M + 1 and M + 2 peaks) to determine the molecular formulas
  – commonly used in inorganic chemistry
- Not useful when the molecular ion peak is not strong or does not appear, and when the isotopic peaks are difficult to locate
  – may give incorrect result(s)
- As the molecules are larger or more complex, both M + 1 and M + 2 peaks becomes significant.

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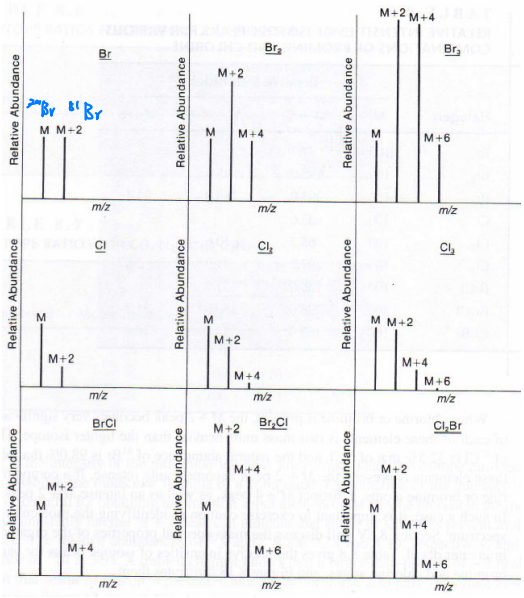
Natural abundances of common elements and their isotopes

Element	Relative Abundance					
Hydrogen	^{1}H	100	^{2}H	0.016		
Carbon	^{12}C	100	^{13}C	1.08		
Nitrogen	^{14}N	100	^{15}N	0.38		
Oxygen	^{16}O	100	^{17}O	0.04	^{18}O	0.20
Fluorine	^{19}F	100				
Silicon	^{28}Si	100	^{29}Si	5.10	^{30}Si	3.35
Phosphorus	^{31}P	100				
Sulfur	^{32}S	100	^{33}S	0.78	^{34}S	4.40
Chlorine	^{35}Cl	100			^{37}Cl	32.5
Bromine	^{79}Br	100			^{81}Br	98.0
Iodine	^{127}I	100				

dep on % of isotope

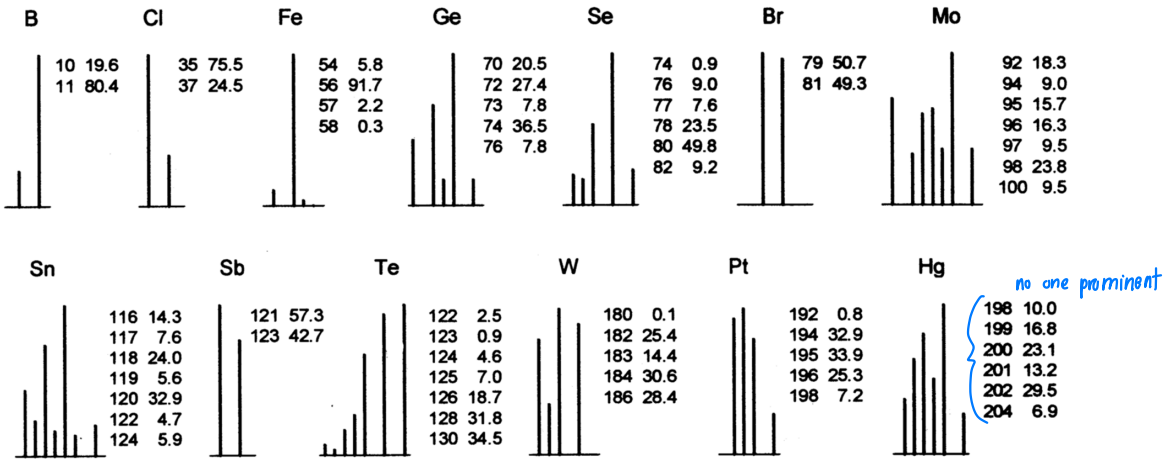
Relative intensities of isotope peaks for halogens

Halogen	Relative Intensities			
	M	M + 2	M + 4	M + 6
Br	100	97.7		
Br_2	100	195.0	95.4	
Br_3	100	293.0	286.0	93.4
Cl	100	32.6		
Cl_2	100	65.3	10.6	
Cl_3	100	97.8	31.9	3.47
BrCl	100	130.0	31.9	
Br_2Cl	100	228.0	159.0	31.2
Cl_2Br	100	163.0	74.4	10.4



* small peaks can be important. predict (integration ratio / pattern when product is successfully synthesized
* inorganic → some products are paramagnetic → x ana ∴ NMR

Natural abundance of some of the common elements



Source of pic.: Alan K. Brisdon, Inorganic Spectroscopic Methods, Oxford, 2001.

Ions-Molecule reaction

- One of the functions of M.S. in organometallic compounds is the <u>possibility</u> to extract the molecular weight => molecular formula

- Some ions can be formed from collision of ions with a molecule, forming ions with greater mass than the molecular ion. <u>These process sometimes might obscure the molecular ion.</u>

- Careful study is needed in some cases:
  - Ions formed from unimolecular process are directly proportional to pressure;
  - Ions formed from bimolecular process are proportional to the square of the pressure : these ions have greater mass than molecular ion. → dilute or reduce P

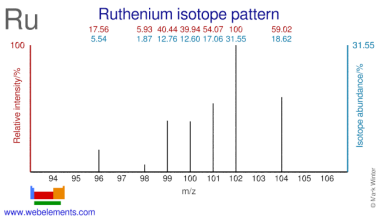
Example: $Cr(CO)_6^+ + Cr(CO)_6 \rightarrow Cr_2(CO)_{10}^+ + 2\ CO$

M^+ M_2^+

Isotopic patterns: isotope & natural abundance

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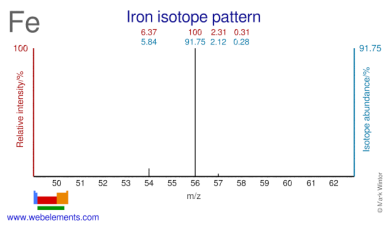
Examples:



Exp 1

Isotope	Mass / Da	Natural abundance (atom %)	Nuclear spin (I)
^{96}Ru	95.907599 (8)	5.54 (14)	0
^{98}Ru	97.905287 (7)	1.87 (3)	0
^{99}Ru	98.9059389 (23)	12.76 (14)	$^5/_2$
^{100}Ru	99.9042192 (24)	12.60 (7)	0
^{101}Ru	100.9055819 (24)	17.06 (2)	$^5/_2$
^{102}Ru	101.9043485 (25)	31.55 (14)	0
^{104}Ru	103.905424 (6)	18.62 (27)	0

only related to NMR. not MS



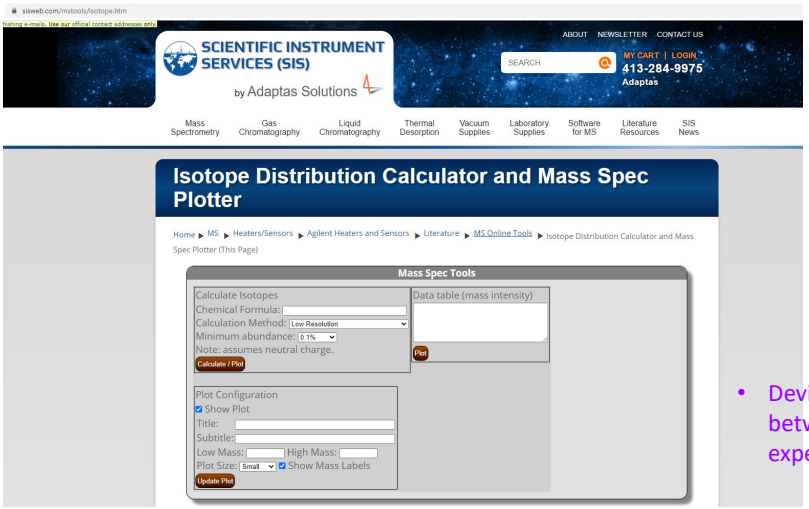
Exp 2

Isotope	Mass / Da	Natural abundance (atom %)	Nuclear spin (I)
^{54}Fe	53.9396127 (15)	5.845 (35)	0
^{56}Fe	55.9349393 (16)	91.754 (36)	0
^{57}Fe	56.9353958 (16)	2.119 (10)	$^1/_2$
^{58}Fe	57.9332773 (16)	0.282 (4)	0

Source: https://www.webelements.com/

Isotope Distribution Calculator and Mass Spec plotter

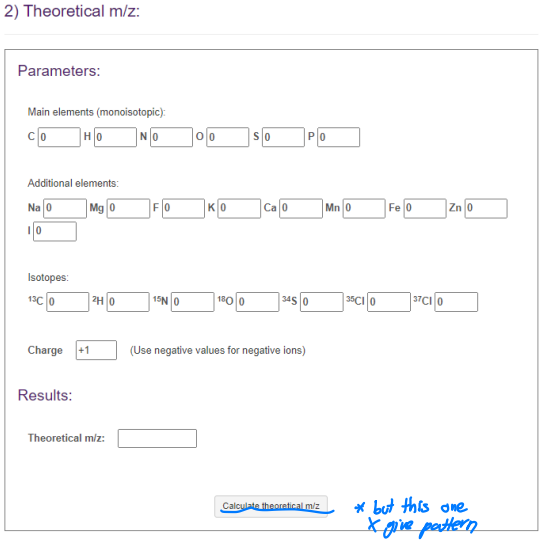
- One of the free online software is from Scientific Instrument Services (SIS)
  (Source: https://www.sisweb.com/mstools/isotope.htm)



- Deviation may be present between calculated one and experimental one.

Other Mass Calculators

Example: https://warwick.ac.uk/fac/sci/chemistry/research/barrow/barrowgroup/calculators/mass_errors/



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INORGANIC Experiment 2

Metallocene: Ferrocene - Synthesis and Characterization

Session I: Synthesis and Microscopic Image of Crude Ferrocene

Session II: Purification and Characterization of Ferrocene

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Iron: oxidation states

- Oxidation states (the most stable states are highlighted in blue):

Group	3	4	5	6	7	8	9	10	11	12
1st row	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
		0	0	0	0	0	0	0	[0]	
			1	1	1	1	1	1	1	[1]
		2	2	2	2	2	2	2	2	2
	3	3	3	3	3	3	3	3	3	
		4	4	4	4	4	4	4	[4]	
			5	5	5					
				6	6	6				
					7					

Reference: C. E. Housecroft, A. G. Sharpe, *Inorganic Chemistry*, 4th ed., Pearson.

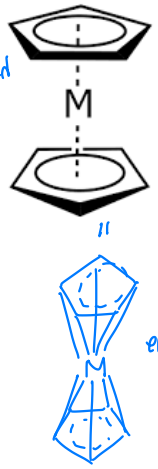
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Metallocene

- Metallocenes are organometallic compounds containing a metal atom (e.g. Fe, Cr, Mn, etc.) "sandwiched" between two aromatic cyclopentadienyl (Cp) rings.
  → Thus, bis(cyclopentadienyl) complexes are called "metallocene" or "sandwich compounds".

- Ferrocene is the first metallocene with the molecular formula $Fe(\eta^5\text{-}C_5H_5)_2$. Its abbreviation is Fc or $Fe(Cp)_2$.

- The synthesis and basic characterization of a ferrocene were reported about seven decades ago by two research groups, T. J. Kealy and P. L. Pauson from Duquesne University in the United States and S. A. Miller, J. A. Tebboth and J. F. Tremaine from the British Oxygen Company, Ltd., in the United Kingdom.



can 6 or 7 mem ring as well.
all 5 C binds to Fe
-ve charge delocalised over 5C.
emphasize that all C interacts w M

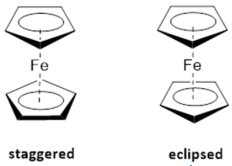
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Ferrocene

- However, the correct sandwich structure, $Fe(Cp)_2$ was reported in 1952 by G. Wilkinson, M. Rosenblum, M. C. Whiting and R. B. Woodward in *Journal of the American Chemical Society* (*JACS*).

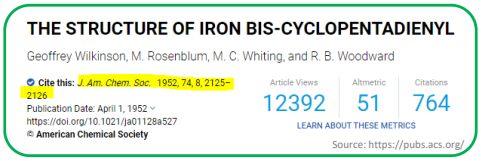
- The independent pioneering work in the chemistry of $Fe(Cp)_2$ led to the joint award of the Nobel Prize in Chemistry to both G. Wilkinson and E. O. Fischer in 1973.

- As a typical metallocene, Fc can exist in two different configurations: eclipsed (D_{5h} symmetry) and staggered (D_{5d} symmetry).



The topics about symmetry and point group are covered in CM3111.

P phy H config (?)


THE STRUCTURE OF IRON BIS-CYCLOPENTADIENYL

Geoffrey Wilkinson, M. Rosenblum, M. C. Whiting, and R. B. Woodward

Cite this: J. Am. Chem. Soc. 1952, 74, 8, 2125–2126

Publication Date: April 1, 1952

https://doi.org/10.1021/ja01128a527

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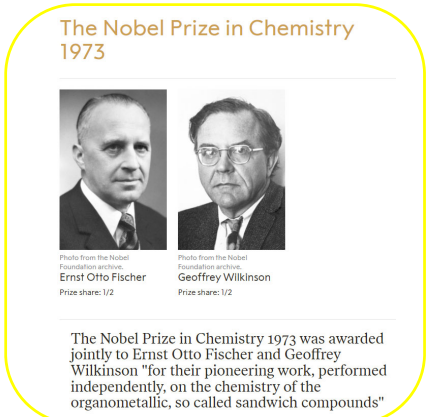

The Nobel Prize in Chemistry 1973

Photo from the Nobel Foundation archive.
Ernst Otto Fischer
Prize share: 1/2

Photo from the Nobel Foundation archive.
Geoffrey Wilkinson
Prize share: 1/2

The Nobel Prize in Chemistry 1973 was awarded jointly to Ernst Otto Fischer and Geoffrey Wilkinson "for their pioneering work, performed independently, on the chemistry of the organometallic, so called sandwich compounds"

References:
1. (a) P. L. Pauson, *J. Organomet. Chem.* 2001, 637-639, 3–6. (b) E. O. Fischer, R. Jira, *J. Organomet. Chem.* 2001, 637-639, 7–12. (c) Myron Rosenblum, *J. Organomet. Chem.* 2001, 637-639, 13–15. (d) P. Štěpnička, *Eur. J. Inorg. Chem.* 2022, 1.
2. (a) G. Wilkinson, M. Rosenblum, M. C. Whiting, R. B. Woodward, *J. Am. Chem. Soc.* 1952, 74, 2125–2126. (b) F. A. Cotton, *J. Organomet. Chem.* 2001, 637-639, 18–26.
3. The Nobel Prize in Chemistry 1973 (https://www.nobelprize.org/prizes/chemistry/1973/summary/)

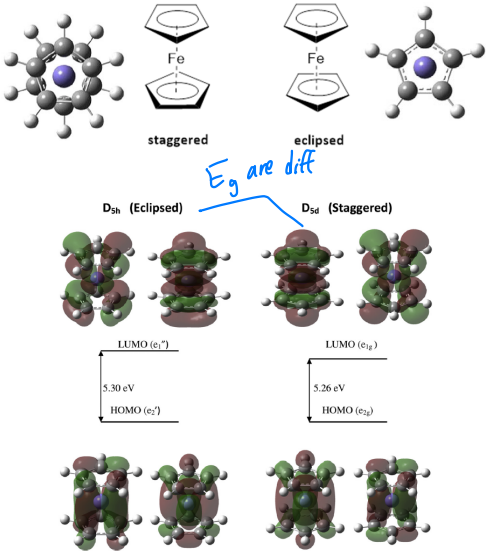
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Ferrocene

- Such two conformers, eclipsed (D_{5h} symmetry) and staggered (D_{5d} symmetry), can be differentiated by a small energy gap.
- It also has been stated in literatures that a staggered conformer dominates experiments in the condensed phase (i.e. solid and liquid states) whereas the eclipsed structure is found in the gas phase.
- The ground electronic state of Fc is a low-spin state as indicated by Gryaznova *et al.* The density-functional theory (DFT) molecular calculations with B3LYP/m6-31G model gave the following configurations and it was published in *J. Organomet. Chem*. **2012.**

D_{5h}, X^1A_2: (core)…$(e_1'')^4(a_1')^2(e_1')^4(e_2')^4(e_2'')^4(a_2'')^2(e_1'')^4(e_1')^4\mathbf{(a_1')^2(e_2')^4(e_1'')^0}$

D_{5d}, X^1A_{1g}: (core)…$(e_{1g})^4(a_{1g})^2(e_{1u})^4(e_{2u})^4(e_{2g})^4(a_{2u})^2(e_{1g})^4(e_{1u})^4\mathbf{(a_{1g})^2(e_{2g})^4(e_{1g})^0}$



References: (1) *J. Am. Chem. Soc.* **1952**, 74, 8, 2125–2126. (2) *J. Organomet. Chem.* **2012**, 713, 51-59

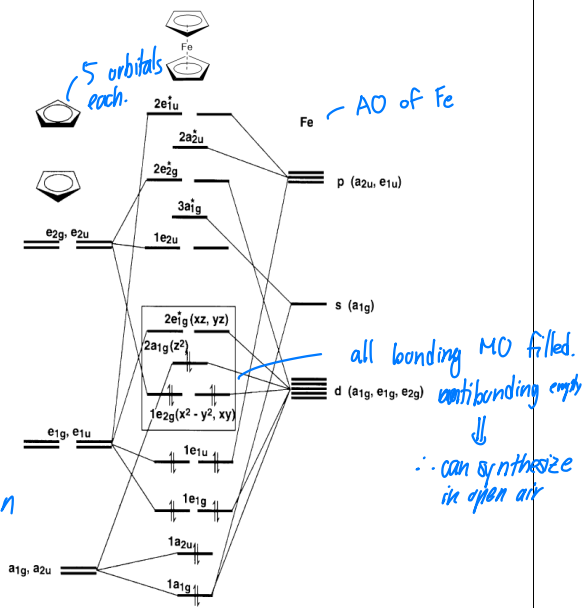
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Ferrocene

- The 10 molecular orbitals (MOs) of the two cyclopentadienyl (Cp) ligands interact with the suitable Fe atomic orbitals (AOs) to give MOs for the $Fe(Cp)_2$.
- In $Fe(Cp)_2$, it contains 18 electrons. Two Cps contribute 10 electrons while Fe metal contributes 8 electrons. These 18 electrons are accommodated in the low lying bonding MOs. No electron is present in any antibonding orbital.
  → $Fe(Cp)_2$ is a stable compound.
- The C-C bonds in $Fe(Cp)_2$ have similar properties as C-C bonds in benzene. Thus, due to such aromatic features, $Fe(Cp)_2$ also exhibits the similar chemical reactions shown by benzene.

The topics about MO diagram are covered in CM3111.

< Qualitative MO diagram for $Fe(Cp)_2$ >



MOs of two Cp molecules | MOs of ferrocene | AOs of Fe

Inorg. Chem. **1999**, *38*, 4861-4867

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Ferrocene: Applications

Biosensors **2021**, 11, 254

biosensors MDPI

Article

An Electrochemical Evaluation of Novel Ferrocene Derivatives for Glutamate and Liver Biomarker Biosensing

since can do reverse coupling

Geok Hong Soon [1], Mary Deasy [1] and Eithne Dempsey [2,*]

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[2] Department of Chemistry, Kathleen Lonsdale Institute for Human Health, Maynooth University, Maynooth, W23 F2H6 Co. Kildare, Ireland
* Correspondence: eithne.dempsey@mu.ie; Tel.: +353-1-4747172

$$\text{Glutamate} + 2Fc^{+} \rightarrow \alpha\text{-ketoglutarate} + NH_3 + 2Fc$$
$$2Fc \rightarrow 2Fc^{+} + 2e^{-}$$

Abstract: Here, we present an evaluation of two new monosubstituted ferrocene (Fc) derivatives, 3-(1H-pyrrol-1-yl)propanamidoferrocene and 1-hydroxy-2-[2-(thiophen-3-yl)-ethylamino]ethylferrocene, as glutamate oxidase mediators, together with their preparation and characterisation. Taking into consideration the influence of the electronic effects of substituents on the redox potentials of the Fc species, two candidates with pyrrole or thiophene moieties were proposed for investigation. Film studies involved potential sweeping in the presence of pyrrole or 3,4-ethylenedioxythiophene monomers resulting in stable electroactive films with % signal loss upon cycling ranging from 1 to 7.82% and surface coverage (Γ) 0.47–1.15 × 10^{-9} mol/cm^2 for films formed under optimal conditions. Construction of a glutamate oxidase modified electrode resulted in second-generation biosensing with the aid of both cyclic voltammetry and hydrodynamic amperometry, resulting in glutamate sensitivity of 0.86–1.28 μA/mM and $K_{m\ (app)}$ values over the range 3.67–5.01 mM. A follow-up enzyme assay for liver biomarker γ-glutamyl transpeptidase realised unmediated and mediated measurement establishing reaction and incubation time investigations and a realising response over <100 U/L γ-glutamyl transpeptidase with a sensitivity of 5 nA/UL^{-1}.

Keywords: glutamate biosensor; liver biomarker; γ-glutamyl transpeptidase; electrochemical sensing; ferrocenes; polymer films

check for updates

Citation: Soon, G.H.; Deasy, M.; Dempsey, E. An Electrochemical Evaluation of Novel Ferrocene Derivatives for Glutamate and Liver Biomarker Biosensing. *Biosensors* **2021**, *11*, 254. https://doi.org/10.3390/bios11080254

Nature Reviews Chemistry, **2017**, 1.

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Published: 06 September 2017

The medicinal chemistry of ferrocene and its derivatives

Malay Patra & Gilles Gasser

Nature Reviews Chemistry **1**, Article number: 0066 (2017) | Cite this article

4686 Accesses | **329** Citations | **11** Altmetric | Metrics

Abstract

Ferrocene derivatives have attracted significant interest as anticancer, antibacterial, antifungal and antiparasitic drug candidates. Discovered in the 1990s, the two most prominent derivatives, ferroquine and ferrocifen, have since been studied extensively for the treatment of malaria and cancer, respectively. The ferrocenyl moiety in these two compounds participates in important metal-specific modes of action that contribute to the overall therapeutic efficacy of the molecules. Although ferroquine is currently in phase II clinical trials and ferrocifen is in preclinical evaluation, no other ferrocene derivative – in fact, no other non-radioactive organometallic compound of any kind – has advanced into clinical trials. This Perspective delineates strategies for the systematic incorporation of ferrocenyl groups into known drugs or drug candidates, with a view to finding new drug leads. In addition, we provide a critical evaluation of the difficulties associated with obtaining the clinical approval that would enable ferrocene-containing molecules to transition from being synthetic curiosities to effective drugs.

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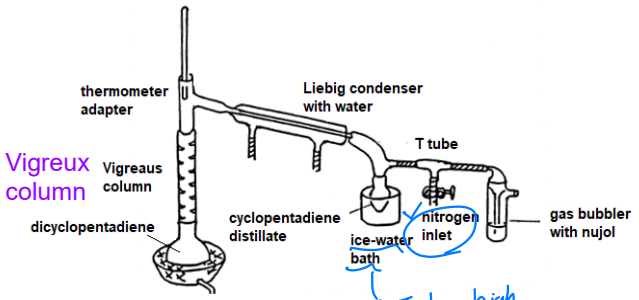
Inorganic experiment 2: Metallocene

Session 0:"Cracking" of dicyclopentadiene *[prepared by lab officers]*

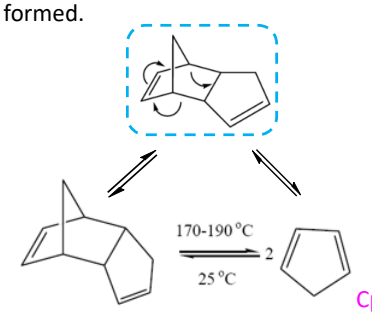
- Upon heating, dicyclopentadiene undergoes cracking (a retro Diels-Alder reaction) to afford two equivalents of free cyclopentadiene (CpH) which can be distilled off once it is formed.

(before deprotonation)

Figure 2. Apparatus used in the cracking of dicyclopentadiene



if T too high, go back to dicyclopentadiene



CpH

dicyclopentadiene	cyclopentadiene (1,3-cyclopentadiene)
density: 0.98 g/mL	density: 0.80 g/mL
b.p.: 170 °C	b.p.: 41 °C
M.W.: 132.20	M.W.: 66.10

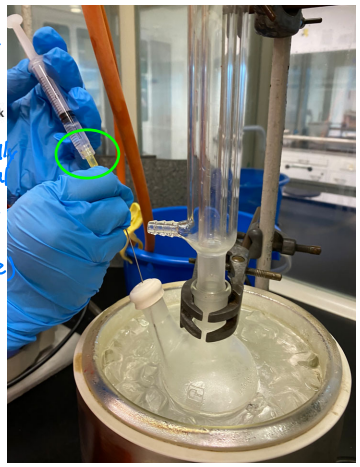
Inorganic experiment 2: Metallocene

Session 0:"Cracking" of dicyclopentadiene [prepared by lab officers]



rubber septum.

usually use metal unless using acid then, must use plastic





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shldn't touch C_5H_5.

1st degas the syringe (so that no air in syringe)
up to 1.5~2mL.
then release N_2 outside
repeat 3x.

cyclopentadiene

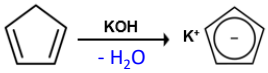
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Inorganic experiment 2: Metallocene

Session I: Synthesis and Microscopic Image of Crude Ferrocene

I-a. Synthesis of Crude Ferrocene

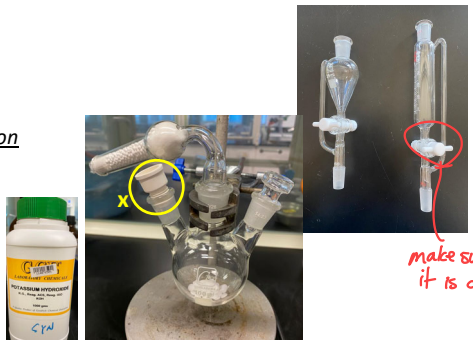
(i) Preparation of potassium cyclopentadienide (also known as cyclopentadienyl) solution



Place a rubber septum in the one of the side necks of a 100 mL three-neck (14/20) round-bottom flask and a glass stopper in another side neck. Quickly weigh and add 4.0 g of finely ground KOH into the round-bottom flask via the middle neck, and cap the flask with a calcium chloride guard tube in the middle neck. Quickly add 12 mL of anhydrous 1,2-dimethoxyethane (DME) from the dispenser into the flask and cap the flask with a calcium chloride guard tube.

base. for de HT

Upon stirring, quickly replace the guard tube with a glass transfer gas adapter connected to the nitrogen gas (N_2) supply and a bubbler on the top, and replace a glass stopper with a pressure-equalizing dropping funnel (DF) [See **Figure 3**]. Turn on the N_2 supply tap slowly and maintain the inert atmosphere (about 1 bubble of N_2 per second) for about 3-5 minutes before proceeding to the next step.



make sure it is closed

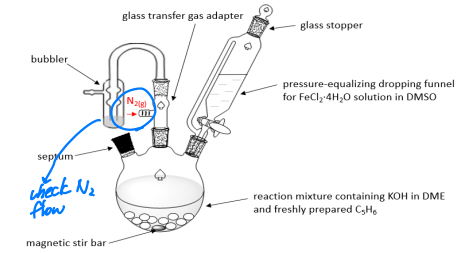


Figure 3. Experimental setup for the synthesis of ferrocene

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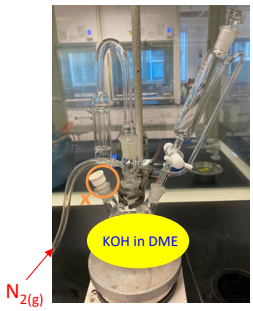
Inorganic experiment 2: Metallocene

Session I: Synthesis and Microscopic Image of Crude Ferrocene

Upon stirring, use a syringe connected to a needle to add 1.0 ml of cyclopentadiene* (prepared by lab officers in Session 0) into the three-neck round-bottom flask via septum. Stir for 10 minutes to obtain a cyclopentadienide solution.

[Warnings: (i) Ensure the solution of iron(II) chloride has been prepared and added into the pressure-equalizing dropping funnel before adding cyclopentadiene monomer into the three-neck round-bottom flask. (ii) Use the needle with care as it is sharp. (iii) Both syringe and needle must be degassed with N_2 first in the receiver flask of the setup of cyclopentadiene, before collecting 1.0 ml of cyclopentadiene. (iv) Inform the lab officer or your GTA or the lecturer before doing such step.]

i.e. Prepare $FeCl_2$ solution first before collecting 1.0 ml of CpH



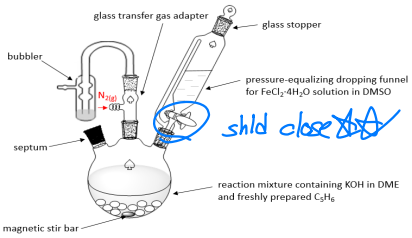


Figure 3. Experimental setup for the synthesis of ferrocene

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Inorganic experiment 2: Metallocene

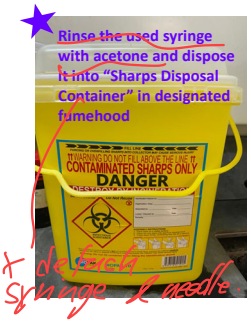
Session I: Synthesis and Microscopic Image of Crude Ferrocene

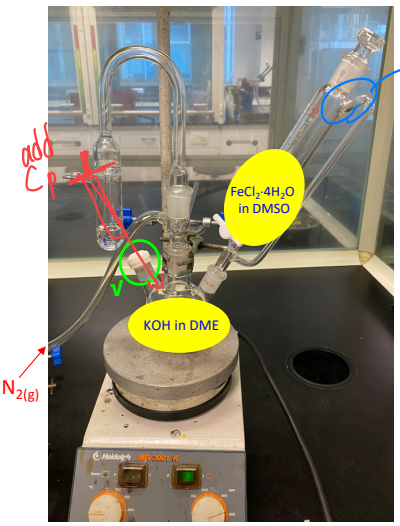
(ii) Preparation of iron(II) chloride solution

Quickly add 1.3 g of $FeCl_2{\cdot}4H_2O$ in a 25 mL round-bottom flask and cap it with calcium chloride guard tube. Add 7 ml of anhydrous dimethylsulfoxide (DMSO) with stirring. Once $FeCl_2{\cdot}4H_2O$ has dissolved into DMSO, use a glass funnel to quickly transfer the solution to the pressure-equalizing dropping funnel (DF). Add ~3.5 ml of anhydrous DMSO to rinse the round-bottom flask and transfer the solution to DF quickly via a glass funnel.

Then collect and add 1.0 ml of CpH with a syringe*

** Degas the syringe in the flask containing CpH distillate*





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Inorganic experiment 2: Metallocene

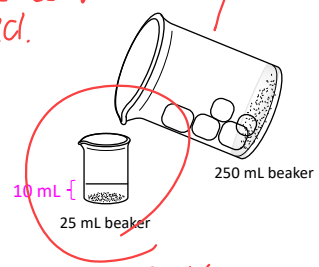
Session I: Synthesis and Microscopic Image of Crude Ferrocene

(iii) Formation of ferrocene

Upon stirring, add the $FeCl_2$ solution slowly (or dropwise at a rate of ~ 20 drops per minute) into the round-bottom flask containing potassium cyclopentadienyl. Continue stirring the mixture for another 30 minutes. Maintain the gentle flow of N_2 in the entire process.

Prepare an ice acidic solution by slow addition^ of 10 mL of concentrated HCl into 30 g of ice in a 250 mL beaker. After 5 minutes, pour 10 mL of the ice acidic solution into a 25 mL beaker. These two beakers of ice acidic solutions will be used in the next step.

[^Warnings: (i) Concentrated HCl is highly corrosive. (ii) Inform the lab officer or your teaching assistant or the lecturer before doing such step.]



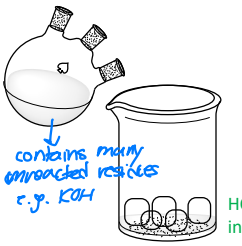
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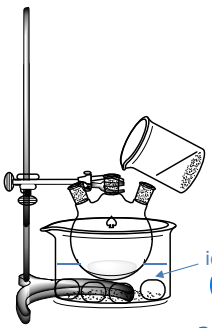
Inorganic experiment 2: Metallocene

Session I: Synthesis and Microscopic Image of Crude Ferrocene

(iii) Formation of ferrocene

For the ferrocene mixture, after 30-minute stirring, stop the N_2 flow. Remove all the glassware connected to the three-neck round-bottom flask. Upon stirring with a glass rod, slowly add the ferrocene reaction mixture from the three-neck round-bottom reaction flask into the 250 mL beaker containing concentrated HCl and ice. Place the three-neck round-bottom reaction flask in an ice-water bath. Add the 10 mL of the ice acidic solution from a 25 mL beaker into the three-neck round-bottom reaction flask upon stirring. Swirl the round-bottom flask and pour all the solution mixture into the 250 mL beaker containing reaction mixture. Rinse the round-bottom flask with 10 mL of deionized water and transfer it into the 250 mL beaker containing reaction mixture. Stir the resulting mixture for 15-20 minutes. Collect the crude ferrocene by suction filtration (with a Büchner funnel). Wash the crude ferrocene with deionized (D.I.) water (3 x 20 mL), and dry it under vacuum (via suction filtration) for ~30 minutes. Record the amount of crude ferrocene obtained and transfer it into a clean sample vial for the next lab session.







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Inorganic experiment 2: Metallocene

Session I: Synthesis and Microscopic Image of Crude Ferrocene

I-b. Microscopic Image of Crude Ferrocene

Record a microscopic image of your crude ferrocene [obtained in Session I (a)] by using the stereomicroscope. With the aid of GTA or lecturer-in-charge, use the lowest magnification ratio of **14x**, select the best view and record the image of your product clearly. Type your matric number on the image captured and send the softcopy of the image to your handphone or NUS email address.

[*Note: The microscopic image obtained must be attached as a part of appendices in the report.*]

[Optional] You can place some crude ferrocene on a watch glass to do this part.



조성 powder (x-crystal)
↓
session 2에서
purify 후, 비교해볼 것.

Inorganic experiment 2: Metallocene

Session I: Synthesis and Microscopic Image of Crude Ferrocene

- At the end of this practical session:

  (i) submit crude ferrocene in a sample vial labelled;

  (ii) obtain microscopic image of your crude ferrocene;

  (iii) record the raw data and observation on your own datasheet.

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Experiment-Based Risk Assessment Form

Name of Department	Department of Chemistry	**Location of Lab**	S5-04
Name of Laboratory	Chemical Synthesis Teaching Lab	**Name of PI (lecturer-in-charge)**	Dr. Edith S. H. Chan
Names of LOs	CM3191 STUDENTS	**Name of Activity/Experiment**	CM3191 Inorganic - Metallocene: Session I - Synthesis and Microscopic Image of Crude Ferrocene

Hazard Identification				Risk Evaluation and Control						
No	Description/Details of Steps in Activity	Hazards	Possible Accident / Ill Health & Persons-at-Risk	Existing Risk Control (Mitigation)	Severity	Likelihood (Probability)	Risk Level	Additional Risk Control	Person Responsible	By (Date)
1	Clean and dry all the glassware (round-bottomed flask, conical flasks, beakers, etc) used in the reactions	Physical hazard: Broken Glassware	Cuts from broken glassware	Visual inspection for any cracks before usage. Proper disposal of broken glassware in Sharps container. Handle & use glass dropper and glassware with care.	2	1	2			
2	Preparation of potassium cyclopentadienide (also known as cyclopentadienyl) solution Place a rubber septum in the one of the side necks of a 100 mL three-neck (14/20) round-bottom flask and a glass stopper in another side neck. Quickly weigh and add 4.0 g of potassium hydroxide (KOH) pellets into the round-bottom flask via the middle neck, and cap the flask with a calcium chloride guard tube in the middle neck.	Chemical hazard: KOH is an irritant and harmful. KOH is corrosive.	Causes irritation to skin. Toxic when ingested. Repeated exposure can be toxic to the body. Spillage of KOH.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use 4. Wash skin thoroughly after handling.	2	1	2			
		Chemical hazard: Calcium chloride is an irritant and harmful.	Calcium chloride can cause eye irritation or damage. Spillage of calcium chloride if the guard tube is broken.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use 4. Wash skin thoroughly after handling.	2	1	2			
3	Quickly add 12 mL of anhydrous 1,2-dimethoxyethane (DME) from the dispenser into the flask and cap the flask with a calcium chloride guard tube.	Chemical and fire hazards: DME is both an irritant and flammable.	DME can cause eye irritation or damage. DME is flammable. Spillage of DME during the transfer process.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use 4. Keep DME away from open or naked flame and a distance away from the hotplate. 5. In case of fire, use dry sand, dry chemical or alcohol resistant foam or water spray or carbon dioxide for extinction.	2	1	2			
		Chemical hazard: Calcium chloride is an irritant and harmful.	Calcium chloride can cause eye irritation or damage. Spillage of calcium chloride if the guard tube is broken.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use 4. Wash skin thoroughly after handling.	1	1	1			

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4	Upon stirring, quickly replace the guard tube with a glass transfer gas adapter connected to the nitrogen gas (N_2) supply and a bubbler on the top, and replace a glass stopper with a pressure-equalizing dropping funnel (DF).	Physical hazard: Broken Glassware	Cuts from broken glassware such as dropping funnel, gas adapter and etc.	Visual inspection for any cracks before usage. Proper disposal of broken glassware in Sharps container. Handle & use glassware with care.	2	1	2			
		Chemical and fire hazards: DME is both an irritant and flammable.	DME can cause eye irritation or damage. DME is flammable. Spillage of DME during the transfer process.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use 4. Keep DME away from open or naked flame and a distance away from the hotplate. 5. In case of fire, use dry sand, dry chemical or alcohol resistant foam or water spray or carbon dioxide for extinction.	2	1	2			
		Chemical hazard: Calcium chloride is an irritant and harmful.	Calcium chloride can cause eye irritation or damage. Spillage of calcium chloride if the guard tube is broken.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use 4. Wash skin thoroughly after handling.	1	1	1			
5	Turn on the N_2 supply tap slowly and maintain the inert atmosphere (about 1 bubble of N_2 per second) for about 3-5 minutes before proceeding to the next step.	Mechanical hazard: pressure induced by N_2(g)	high pressure generated in the reaction flask will cause chemical spillage and the breakage of glassware	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use 4. Adjust the pressure of the N_2(g) via the gas inlet.	2	1	2			
6	Upon stirring, use a syringe connected to a needle to add 1.0 ml of cyclopentadiene (prepared by lab officers in Session 0) into the three-neck round-bottom flask (containing DME and KOH) via septum. Stir for 10 minutes to obtain a cyclopentadienide solution.	Chemical and fire hazards: Cyclopentadiene is both an irritant and flammable	Cyclopentadiene can cause eye irritation or damage. cyclopentadiene is flammable. Spillage of cyclopentadiene during the transfer process.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use 4. Keep cyclopentadiene away from open or naked flame and a distance away from the hotplate. 5. In case of fire, use dry sand, dry chemical or alcohol resistant foam or water spray or carbon dioxide for extinction.	2	1	2			
		Chemical hazard: KOH is an irritant and harmful. KOH is corrosive.	Causes irritation to skin. Toxic when ingested. Repeated exposure can be toxic to the body. Spillage of KOH.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use 4. Wash skin thoroughly after handling.	2	1	2			

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		Chemical and fire hazards: DME is both an irritant and flammable.	DME can cause eye irritation or damage. DME is flammable. Spillage of DME during the transfer process.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use. 4. Keep DME away from open or naked flame and a distance away from the hotplate. 5. In case of fire, use dry sand, dry chemical or alcohol resistant foam or water spray or carbon dioxide for extinction.	2	1	2			
		Physical hazard: Broken glassware	Cuts from broken glassware	Visual inspection for any cracks before usage. Proper disposal of broken glassware in Sharps container. Handle & use glassware with care.	2	1	2			
		Electrical hazard: Usage of hotplate, exposed wire of the cable	Electric shock if used improperly.	1. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 2. Visually inspect apparatus for cracks before use. 3. In case of fire, use dry sand, dry chemical or alcohol resistant foam or water spray or carbon dioxide for extinction.	2	1	2			
7	Preparation of iron(II) chloride solution Quickly add 1.3 g of FeCl2·4H2O in a 25 mL round-bottom flask and cap it with calcium chloride guard tube. Add 7 ml of anhydrous dimethylsulfoxide (DMSO) with stirring. Once FeCl2·4H2O has dissolved into DMSO, use a glass funnel to quickly transfer the solution to the pressure-equalizing dropping funnel (DF). Add 3.5 ml of anhydrous DMSO to rinse the round-bottom flask and transfer the solution to DF quickly via a glass funnel.	Chemical hazard: FeCl2·4H2O is corrosive, harmful and irritating.	Causes irritation or damage to eye and skin. Toxic when ingested.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Wash skin thoroughly after handling.	2	1	2			
		Chemical hazard: DMSO is harmful.	Rapidly absorbed through skin.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Wash skin thoroughly after handling.	2	1	2			
8	Formation of ferrocene Upon stirring, add the FeCl2 solution dropwise at a rate of ~ 20 drops per minute into the round-bottom flask containing potassium cyclopentadienyl. Continue stirring the mixture for another 30 minutes. Maintain the gentle flow of N2 in the entire process.	Chemical hazard: FeCl2·4H2O is corrosive, harmful and irritating.	Causes irritation or damage to eye and skin. Toxic when ingested.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Wash skin thoroughly after handling.	2	1	2			
		Chemical hazard: DMSO is harmful.	Rapidly absorbed through skin.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Wash skin thoroughly after handling.	2	1	2			

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		Chemical and fire hazards: Cyclopentadiene is both an irritant and flammable	Cyclopentadiene can cause eye irritation or damage. cyclopentadiene is flammable. Spillage of cyclopentadiene during the transfer process.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use 4. Keep cyclopentadiene away from open or naked flame and a distance away from the hotplate 5. In case of fire, use dry sand, dry chemical or alcohol resistant foam or water spray or carbon dioxide for extinction.	2	1	2			
		Chemical hazard: KOH is an irritant and harmful. KOH is corrosive.	Causes irritation to skin. Toxic when ingested. Repeated exposure can be toxic to the body. Spillage of KOH.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use 4. Wash skin thoroughly after handling.	2	1	2			
		Chemical and fire hazards: DME is both an irritant and flammable.	DME can cause eye irritation or damage. DME is flammable. Spillage of DME during the transfer process.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use 4. Keep DME away from open or naked flame and a distance away from the hotplate. 5. In case of fire, use dry sand, dry chemical or alcohol resistant foam or water spray or carbon dioxide for extinction.	2	1	2			
		Mechanical hazard: pressure induced by N2(g)	high pressure generated in the reaction flask will cause chemical spillage and the breakage of glassware	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use 4. Adjust the pressure of the N2(g) via the gas inlet.	2	1	2			
9	Prepare an ice acidic solution by slow addition of 10 mL of concentrated HCl into 30 g of ice in a 250 mL beaker. After 5 minutes, pour 10 mL of the ice acidic solution into a 25 mL beaker. These two beakers of ice acidic solutions will be used in the next step.	Chemical hazard: HCl is an irritant and harmful. HCl is corrosive.	Causes irritation to skin. Toxic when ingested. Repeated exposure can be toxic to the body. Spillage of HCl.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use 4. Wash skin thoroughly after handling.	2	1	2			
		Physical hazard: addition of concentrated HCl into ice	Spillage of concentrated HCl	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Add concentrated HCl slowly into ice in the well centilated fumehood. 4. Wash skin thoroughly after handling.	2	1	2			

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10	After 30-minute stirring, stop the N2 flow. Remove all the glassware connected to the three-neck round - bottom flask. Upon stirring with a glass rod, slowly add the ferrocene reaction mixture from the three-neck round-bottom reaction flask into the 250 mL beaker containing concentrated HCl and ice. Place the three-neck round-bottom reaction flask in an ice-water bath. Add the 10 mL of the ice acidic solution from a 25 mL beaker into the three-neck round-bottom reaction flask upon stirring. Swirl the round-bottom flask and pour all the solution mixture into the 250 mL beaker containing reaction mixture. Rinse the round-bottom flask with 10 mL of deionized water and transfer it into the 250 mL beaker containing reaction mixture.	Chemical hazard: HCl is an irritant and harmful. HCl is corrosive.	Causes irritation to skin. Toxic when ingested. Repeated exposure can be toxic to the body. Spillage of HCl.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use 4. Wash skin thoroughly after handling.	2	1	2			
		Physical hazard: pouring the reaction mixture containing KOH into cold HCl solution, or addition of cold HCl solution into the reaction mixture containing KOH	Spillage of HCl or KOH or ferrocene	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Add the chemicals from one reaction mixture to another reaction solution slowly. 4. Wash skin thoroughly after handling.	2	1	2			
		Chemical hazard: KOH is an irritant and harmful. KOH is corrosive.	Causes irritation to skin. Toxic when ingested. Repeated exposure can be toxic to the body. Spillage of KOH.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use 4. Wash skin thoroughly after handling.	2	1	2			
		Chemical and fire hazards: DME is both an irritant and flammable.	DME can cause eye irritation or damage. DME is flammable. Spillage of DME during the transfer process.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use 4. Keep DME away from open or naked flame and a distance away from the hotplate. 5. In case of fire, use dry sand, dry chemical or alcohol resistant foam or water spray or carbon dioxide for extinction.	2	1	2			
		Chemical and fire hazards: Ferrocene is harmful, toxic, irritating and flammable.	Ferrocene can cause eye irritation or damage, and causes aquatic hazard in an environment. Ferrocene is flammable. Spillage of Ferrocene during the transfer process.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Discard ferrocene into proper waste bottle. 4. Keep ferrocene away from open or naked flame and a distance away from the hotplate. 5. In case of fire, use dry sand, dry chemical or alcohol resistant foam or water spray or carbon dioxide for extinction. 6. Wash skin thoroughly after handling.	2	1	2			

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11	Stir the resulting mixture for 15-20 minutes. Collect the crude ferrocene by suction filtration (with a Büchner funnel). Wash the crude ferrocene with deionized (D.I.) water (3 x 20 mL), and dry it under vacuum for 30 minutes. Record the amount of crude ferrocene obtained and transfer it into a clean sample vial for the next lab session.	Chemical and fire hazards: Ferrocene is harmful, toxic, irritating and flammable.	Ferrocene can cause eye irritation or damage, and causes aquatic hazard in an environment. Ferrocene is flammable. Spillage of Ferrocene during the transfer process.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Discard ferrocene into proper waste bottle. 4. Keep ferrocene away from open or naked flame and a distance away from the hotplate. 5. In case of fire, use dry sand, dry chemical or alcohol resistant foam or water spray or carbon dioxide for extinction. 6. Wash skin thoroughly after handling.	2	1	2			
		Mechanical & Physical hazards: Usage of set-up for suction filtration	Glassware might topple down when the vacuum is used and could break, resulting in cuts.	1. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 2. Visually inspect apparatus for cracks before use. 3. Clamp the apparatus properly before usage.	2	1	2			
12	Record a microscopic image of your crude ferrocene by using the stereomicroscope	Chemical and fire hazards: Ferrocene is harmful, toxic, irritating and flammable.	Ferrocene can cause eye irritation or damage, and causes aquatic hazard in an environment. Ferrocene is flammable. Spillage of Ferrocene during the transfer process.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Discard ferrocene into proper waste bottle. 4. Keep ferrocene away from open or naked flame and a distance away from the hotplate. 5. In case of fire, use dry sand, dry chemical or alcohol resistant foam or water spray or carbon dioxide for extinction. 6. Wash skin thoroughly after handling.	2	1	2			
		Physical hazard / Human factor: Spillage of chemicals (ferrocene)	Spillage of chemicals during the record of microscopic image	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Handle the chemicals with care.	2	1	2			
		Electrical hazard: exposed wire of the cable and/or broken plug or socket for the stereomicroscope	Electric shock if used improperly	1. Check the cable, plug and socket before use. 2. In case of fire, use dry sand, dry chemical or alcohol resistant foam or water spray or carbon dioxide for extinction. 3. Refer the instrumental operation manual.	2	1	2			
13	Washing of glasswares with acetone after experiment is completed	Chemical and fire hazards: Acetone and methanol are irritants and are flammable.	Can cause irritation to skin and eyes. Flammable substance that can start a fire.	1.Wash the apparatus in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Avoid placing acteone and methanol near any open or naked flame 4. In case of fire, use dry sand, dry chemical or alcohol resistant foam or water spray or carbon dioxide for extinction.	2	1	2			

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		Physical hazard: Broken glasswares	Cuts and injury from broken glassware	1.Wash the apparatus in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Handle the glasswares with care to avoid dropping them 4. Dispose the broken glassware into the appropriate Sharps bin	2	1	2			

Conducted By

Name _____

Signature _____

date _____

Approved By

Name Dr. Edith Chan

Signature _____

Approval date _____ Next Revision date _____ (Maximum 3 years)

Reminder

Experiment-Based Risk Assessment Form

Name of Department _____ Name and Location of Lab _____

Name of Laboratory _____ Name of PI (lecturer-in-charge) _____

Name of Student _____ Name of Activity/Experiment _____

Hazard Identification				Risk Evaluation & Control						
No	Description / Details of Steps in Activity	Hazard(s)	Possible Accident(s)	Existing Risk Control (Mitigation)	Severity	Likelihood (probability)	Risk Level	Additional Risk Control	Person Responsible	By (Date)

- **Submit your filled risk assessment for inorganic expt. 2 (session II) *by* 10:00 a.m. on 16-Sep-2024 (Mon.; Week 6) via Canvas-Assignments [5% CA]. Penalty will be applied for any late submission.**

3 of Edit 0 additional risk control & person responsibles first

Conducted By _____

Name _____

Signature _____

date _____

Approved By _____

Name _____

Signature _____

Approval date _____ Next Revision date _____ (Maximum 3 years)

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Inorganic experiment 2: Metallocene

Session II: Purification and Characterization of Ferrocene

Record the weights of the glass petri dish covers (bigger size) first with an analytical balance. Spread the crude ferrocene on a glass petri dish base (smaller size) and cover it with the glass petri dish cover (bigger size). Purify the crude ferrocene by sublimation using the setup depicted in **Figure 4**. Slowly raise the temperature (< 150 °C) until the ferrocene sublimes to the dish cover. The sublimation will proceed slowly. Cooling the dish cover by placing a 400 mL beaker filled with ice water on top of it to facilitate the sublimation [see **Figure 4**]. Remove the glass petri dish cover (bigger size) from the heat, allow it to cool and recover the sublimed ferrocene [Note: This procedure may be repeated twice (or more) until all the purified ferrocene has been obtained.]. Allow the petri dish cover filled with sublimed ferrocene to cool to room temperature. Record the weight(s) of the petri dish covers filled with ferrocene sublimed. Transfer the ferrocene sublimed from the petri dish cover(s) into a clean transparent sample plastic bag.

[Note: The calculation of % yield of the purified ferrocene obtained must be shown in the report.]

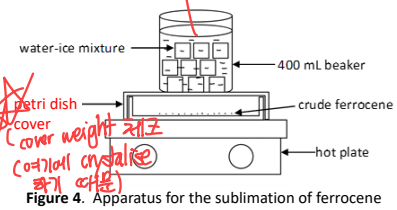
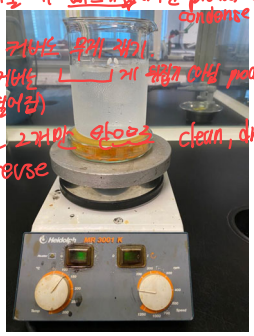


Figure 4. Apparatus for the sublimation of ferrocene



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Inorganic experiment 2: Metallocene

Session II: Purification and Characterization of Ferrocene

(i) Record of microscopic image of the purified ferrocene

Obtain a microscopic image of your purified ferrocene by using the stereomicroscope *(refer to the stereomicroscope instruction manual in the teaching lab.)*. With the aid of GTA or lecturer-in-charge, use the lowest magnification ratio of **14x**, select the best view and record the image of your purified product clearly. Type your matric number on the image captured and send the softcopy of the image to your handphone or NUS email address.

[*Note: The microscopic image obtained must be attached as a part of appendices in the report.*]

(ii) Measurement of magnetic susceptibility

Use agate mortar and pestle to grind 0.09-0.12 g of your purified ferrocene into fine powder, and follow the magnetic susceptibility balance instruction manual [Refer to **Appendix**] to measure the magnetic susceptibility of your purified ferrocene. → 2 values: Volume susceptibility (χ_v) and Mass susceptibility (χ_m)

*[Importance: Take pictures to record all the values obtained from the magnetic susceptibility balance in each step stated in the instruction manual (see **Appendix**) and attach them as a part of appendices in the lab report.]*



Magnetic Susceptibility Balance
(Model: Sherwood MSB AUTO)

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Inorganic experiment 2: Metallocene

Session II: Purification and Characterization of Ferrocene

(ii) Measurement of magnetic susceptibility



Appendix (lab manual)

Operation of Magnetic Susceptibility Balance (Model: Sherwood MSB AUTO)



1. Switch on the instrument.
2. Wait until a zero value is obtained (-0.000E-5Z). Take a picture on the control unit for such reading.
3. Weight an empty sample tube. Place an empty sample tube of known weight to the detector and obtained a reading.
4. Press the "Tare" function, and observed the display to ensure the numerical value of zero is displayed.
5. Remove the sample tube and introduce the sample. Measure the weight and length of the sample packed, with an analytical balance and a ruler. (Note: The length must be between 15.0 and 25.0 mm.)
6. Press the "Length" function button and observe the display.
   Length = _ _._ mm
7. Enter the length by pressing the appropriate three number keys in order. Take a picture on the screen of the control unit before entering the last number.
8. Following the input of the appropriate length, the display indicates "processing" and then returns to the Xv-Tare display. Press the "Weight" function button and observe the display. Take a picture on the screen of the control unit before entering the last number.
   Weight = 0._ _ _ _ gm
   (Note: The weight value between 0.0010 grams and 0.9999 grams are expected by the Auto.)
9. The display will return to the Xv-Tare display, press the "Mag Sus" button and observe the display.

Take a picture on the control unit for each of the following readings.

(i) Volume Suscept → x.xxx E-xV Tare
(ii) Mass Suscept → x.xxx E-xM Tare

[Note: M – Mass susceptibility; V – Volume susceptibility]

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Inorganic experiment 2: Metallocene

Session II: Purification and Characterization of Ferrocene

(iii) UV-visible Spectroscopic Characterization of the purified ferrocene

Prepare the standard solution of purified ferrocene (35-40 mg, weighed accurately by the analytical balance) in methanol in a 25 mL volumetric flask and record its UV-Vis spectrum (300–700 nm). By doing proper dilutions in the fume cupboard, collect the spectra showing weak and strong peaks (A = 0.1–1.0). Determine the molar extinction coefficients ε ($M^{-1}cm^{-1}$) for these peaks obtained from the selected spectra and report them in the proforma lab report.

[Note: All the ε calculation involved must be handwritten clearly on the selected UV-vis spectra, and ALL the UV-vis spectra obtained must be attached as a part of appendices in the short lab report.]

(iv) Infrared Characterization of the purified ferrocene

Obtain the solid IR spectrum of the purified ferrocene by *either* ATR-IR *or* KBr disc. Record the spectra within the range 500–4000 cm^{-1}.

[Note: The IR characterization method chosen must be recorded on your datasheet and the IR spectra obtained must be attached as a part of appendices in the short lab report.]

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Inorganic experiment 2: Metallocene

Session II: Purification and Characterization of Ferrocene

(v) **^{1}H-NMR Characterization of the purified ferrocene** → **Sample preparation only** (S/N no. must be stated on the label)

Transfer the purified ferrocene (~ 1 mg) into the NMR tube containing 0.6 mL of $CDCl_3$ Label the sample tube clearly and submit it for ^{1}H-NMR.

[Note: ALL the NMR spectra obtained must be attached as a part of appendices in the short lab report.]

(vi) **ESI-MS Characterization of the purified ferrocene** → **Sample preparation only** (S/N no. must be stated on the label)

Transfer the purified ferrocene (1-2 mg) into a sample tube. Label the sample tube clearly and submit it for Electrospray Ionization Mass Spectrometry (ESI-MS) analysis.

[Note: ALL the ESI-MS spectra obtained must be attached as a part of appendices in the short lab report.]

Inorganic experiment 2: Metallocene

Session II: Purification and Characterization of Ferrocene

- At the end of this practical session:

(i) submit your ferrocene in a sample vial labelled and purified ferrocene in a sample plastic bag;

(ii) obtain microscopic image of your purified ferrocene;

(iii) measure the magnetic susceptibility your purified ferrocene and take pictures for the measurement process;

(iv) obtain IR and UV-vis spectra of purified ferrocene

(v) submit your purified ferrocene sample for ESI-MS;

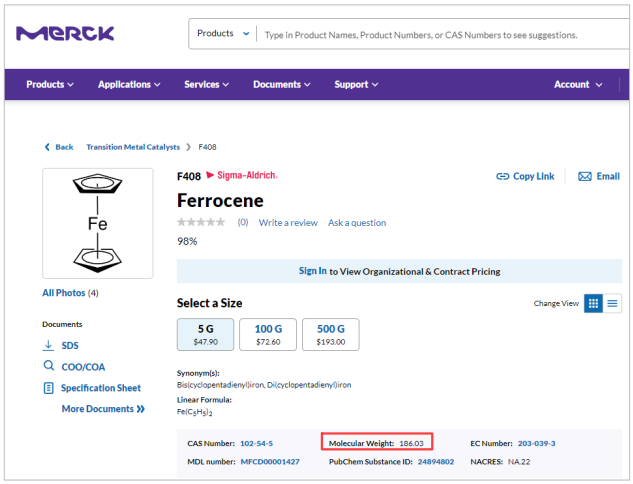
(vi) submit your purified ferrocene sample for 1H-NMR;

(vii) record the raw data and observation on your own datasheet.

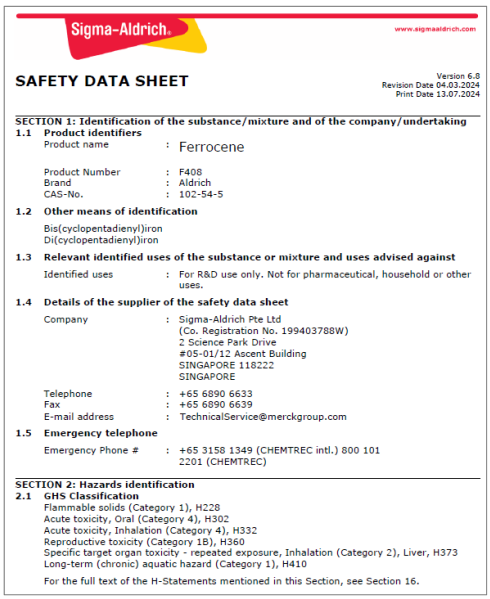
Reminder: Take note on the short report requirement for inorganic experiment 2 [please refer to the Inorganic notes p.23-25].

Inorganic experiment 2: Metallocene

Session II: Purification and Characterization of Ferrocene



https://www.sigmaaldrich.com/SG/en/product/aldrich/f408



https://www.sigmaaldrich.com/SG/en/sds/aldrich/f408

Ferrocene: Purification

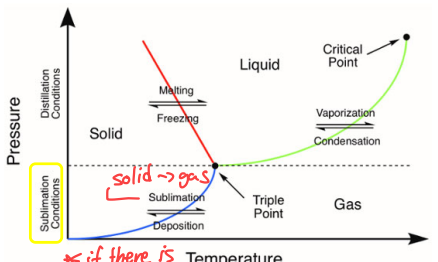
(liq, solid & gas all eq)

below triple point

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- Ferrocene is purified with sublimation at < 150°C.
- Sublimation is a direct phase transition of a solid state into a gaseous state without liquid state.
- Sublimation takes place at temperatures and pressures lower than that of the substance's triple point.
- Triple point: The temperature and pressure at which the solid, liquid and gaseous states of a substance can coexist in a stable equilibrium.
- The non-volatile impurities are left behind while the sublimed compound is then deposited as a solid on a cold surface.
- Ferrocene can be also purified with recrystallization in methanol.

Generic phase diagram



Source of picture: https://www.jove.com/

* if there is impurity, stay on base

* H_2O may also condense on cover

Microscopic image of ferrocene crystals



Inorganic experiment 2: Metallocene

Session I-b & II-b-i. Microscopic Images of Crude Ferrocene & the Purified Ferrocene

Optical Examination -Stereomicroscopy — no quanti value, just quali observation.

- reveals the bulk sample features in a microscopic level, e.g. crystal size, shape, color and bulk homogeneity
- using the lowest magnification ratio of 14x, select the best view and take ONE picture of each crude/purified ferrocene placed on the glass petri dish cover or in a sample plastic bag or a watch glass
- adjust the focus or depth of field of the view while you view the samples
- sign the record sheet and submit your ferrocene samples finally, before you leave the lab!
- label each image with your matric no.
- send each picture to your phone or NUS email account



Courtesy of Dr. Stephen Chui, NUS